

Midterm Exam (part 1) - Computational Physics I

NAME: Juan Daniel Vasconez

SCORE: 7.3
8

Date: Thursday 3 April 2025 Duration: 45 minutes

Credits: 8 points (4 questions) Type of evaluation: MT

Please provide concise answers to the following items:

1. (2 points) Python Input/Output

- Describe 2 distinct methods suitable for handling tabulated data input and output in Python.
- Explain one advantage and one disadvantage of each method.

a) **Open:** It's a native python feature. That has different modes for opening a file, (r, w, a, rt) and depends on for loops to make changes or extract data.

pandas: is a third-party library which helps us create dataframes and handle data as a spreadsheet, it's more user friendly.

b) **Open:**

A → Present natively, constant version.

D → A bit convoluted, no user friendly.

pandas

A → Simpler to use and more intuitive.

D → can suffer deprecations, and a bit slow for big data.

2. (2 points) Data fitting methods for beginners

- List the main steps to carry out a meaningful regression using empirical data in Python.
- Explain the difference between the least-squares (LM) and the χ^2 data fitting methods.

a) → Open data

→ Inspect the data

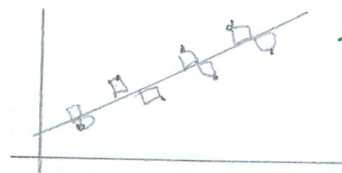
→ Check linearity or try to linearize the problem with any variable changes.

→ Propose a physically motivated model, and a educated guess.

→ Fit the model and get the free parameters depending on the ansatz.

b)

LM: Here we will minimize a model direct functional, trying to reduce the mean area of imaginary squares on the points.



χ^2 : Is a different Functional that takes into account each point dispersion and error to provide a better fit.

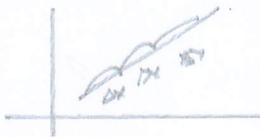
$$F(x, y) = \frac{(y_i - f_i(x, y))^2}{\sigma_i^2}$$

points → y_i
model → $f_i(x, y)$
 $\sigma_i^2 \rightarrow$ SD.

3. (2 points) Numerical differentiation

- a. Explain how the finite-difference methods for calculating derivatives work.
b. Provide the mathematical definition of a central-difference scheme.

a) Finite differences method for derivatives come from the definition of the derivative itself, taking points and its neighbours to create a difference between them and divide it by the Δx we take, similar to the derivative definition using h .

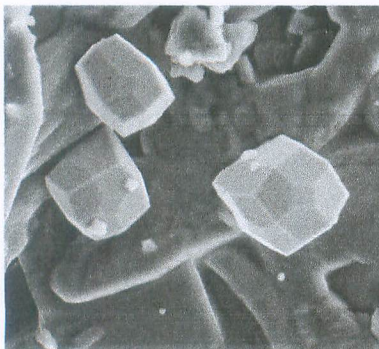


b)

$$\frac{df}{dx} = \frac{f_{i-1} - 2f_i + f_{i+1}}{2\Delta x} + O(\Delta x^2)$$

where f_i is the step and takes the previous step or the next step to create difference. This method on the edges should fall to a forward or backward derivative.

4. (2 points) Image processing



Imagine you obtain this photograph of iron crystals from a scanning electron microscope (credits: NASA/JSC), and you are asked to isolate the more prominent crystals from the background and from the rest of the image. Design and sketch a suitable algorithm workflow to achieve this goal in Python.

